

MOCK INTERVIEW SCRIPT (WITH FOLLOW-UPS)

ROUND 1: INTRO + FUNDAMENTALS

Q1. Tell me about a Machine Learning project you worked on.

Expected Answer Structure (STAR):

- **Problem** – What was the business problem?
- **Data** – What data did you use?
- **Approach** – Model selection + features
- **Result** – Metrics + business impact

Follow-ups:

- Why did you choose that algorithm?
- What alternatives did you consider?
- What would you improve if given more time?

Q2. How do you approach any ML problem?

Expected Answer:

1. Problem understanding
2. Data collection & cleaning
3. EDA
4. Feature engineering
5. Model selection
6. Evaluation
7. Deployment

Follow-ups:

- Where do most projects fail?
- How do you handle unclear problem statements?

Q3. Explain Bias-Variance Tradeoff

Follow-ups:

- How do you detect high variance in real projects?
 - Give 2 practical techniques to reduce variance.
-

ROUND 2: MODELING + EVALUATION

Q4. You built a classification model with 95% accuracy. Is it good?

Expected Answer:

- Depends on dataset balance
- Check confusion matrix, precision, recall

Follow-ups:

- What if dataset is imbalanced?
 - Which metric would you prioritize in fraud detection?
-

Q5. How do you handle missing data?

Follow-ups:

- When would you drop vs impute?
 - Can missing data itself be useful?
-

Q6. Explain Overfitting in a real-world context

Follow-ups:

- How do you detect it?
 - What are 3 ways to prevent it?
-

ROUND 3: SYSTEM THINKING

Q7. How would you deploy an ML model?

Expected Answer:

- Train → Serialize → API → Deployment → Monitoring

Follow-ups:

- How do you monitor model performance?
- What is model drift?

Q8. How do you handle real-time vs batch ML systems?

Follow-ups:

- When would you use streaming (Kafka)?
- Tradeoffs between latency vs accuracy

ROUND 4: ADVANCED / SENIOR

Q9. How would you debug a model that suddenly drops performance?

Expected Answer:

- Data drift
- Feature issues
- Pipeline bugs
- Concept drift

Follow-ups:

- How do you detect drift?
- Tools you would use?

Q10. How do you ensure scalability of ML systems?

Expected Answer:

- Distributed computing (Spark)
 - Model optimization
 - Microservices
 - Caching
-
-

CASE STUDY-BASED ML ROUNDS (AMAZON-STYLE)

HOW INTERVIEWERS EVALUATE YOU

- Problem structuring
- Business understanding
- ML application
- Communication

(Case studies simulate real-world ML problems and test applied thinking rather than theory)

CASE 1: E-COMMERCE RECOMMENDATION SYSTEM

Problem:

Design a recommendation system for an e-commerce platform.

Expected Approach:

1. Clarifying Questions

- What type of recommendations? (products, bundles)
- Real-time or batch?
- Cold start problem?

2. Data Required

- User behavior (clicks, purchases)
- Product metadata
- Ratings

3. Approach

- Collaborative filtering
- Content-based filtering
- Hybrid models

4. Evaluation Metrics

- Precision@K
- Recall@K

5. Challenges

- Cold start
- Scalability
- Data sparsity

Follow-ups:

- How would you handle new users?
- How would you scale to millions of users?

CASE 2: FRAUD DETECTION SYSTEM

Problem:

Detect fraudulent transactions in banking.

Expected Approach:

1. Key Challenges

- Highly imbalanced data
- Real-time predictions

2. Data

- Transaction history
- User behavior
- Location patterns

3. Approach

- Classification models (XGBoost, RF)
- Anomaly detection

4. Metrics

- Precision, Recall (NOT accuracy)

5. Production Considerations

- Latency
- False positives impact

Follow-ups:

- What happens if false positives are high?
 - How would you reduce them?
-

CASE 3: CUSTOMER CHURN PREDICTION

Problem:

Predict which users will churn.

Expected Approach:

1. Define churn clearly

- No login? No purchase?

2. Data

- User activity
- Subscription history

3. Feature Engineering

- Recency, frequency
- Behavioral patterns

4. Model

- Logistic Regression / Tree-based

5. Business Impact

- Retention campaigns

Follow-ups:

- How would you make this actionable?
 - How do you explain model predictions to business?
-
-

GOLDEN FRAMEWORK (CANDIDATE SHOULD FOLLOW)

Whenever given ANY case:

1. Clarify problem
2. Define success metrics
3. Identify data
4. Choose approach
5. Evaluate
6. Discuss trade-offs
7. Production considerations

(Strong candidates explicitly structure answers this way in ML case interviews)

MACHINE LEARNING – INTERVIEW QUESTIONS & ANSWERS

1. BASIC LEVEL

1. What is Machine Learning?

Answer:

Machine Learning is a subset of Artificial Intelligence where systems learn patterns from data and make predictions or decisions without being explicitly programmed.

2. Types of Machine Learning

Answer:

- **Supervised Learning** – Uses labeled data (e.g., regression, classification)
 - **Unsupervised Learning** – Finds patterns in unlabeled data (e.g., clustering)
 - **Reinforcement Learning** – Learns via rewards/penalties from environment
-

3. Difference between AI, ML, and Deep Learning

Answer:

- **AI** – Broad concept of machines mimicking human intelligence
 - **ML** – Subset of AI that learns from data
 - **Deep Learning** – Uses neural networks with multiple layers
-

4. What is Overfitting and Underfitting?

Answer:

- **Overfitting** – Model learns noise; high training accuracy, low test accuracy
- **Underfitting** – Model too simple; poor performance on both

Fix:

- Overfitting → Regularization, pruning, cross-validation
 - Underfitting → Increase complexity, more features
-

5. What is Bias-Variance Tradeoff?

Answer:

- **Bias** → Error due to simplicity
 - **Variance** → Error due to complexity
 - Goal → Balance both for good generalization
-

6. What is Gradient Descent?

Answer:

An optimization algorithm used to minimize loss function by iteratively updating parameters in the direction of negative gradient.

7. What is a Loss Function vs Cost Function?

Answer:

- **Loss Function** – Error for a single data point
 - **Cost Function** – Aggregate loss over entire dataset
-
-

2. INTERMEDIATE LEVEL

8. What are Evaluation Metrics for Classification?

Answer:

- Accuracy
- Precision
- Recall
- F1 Score

Formula:

$$F1 = 2TP / (2TP + FP + FN)$$

9. Precision vs Recall

Answer:

- **Precision** → Out of predicted positives, how many are correct
 - **Recall** → Out of actual positives, how many are identified
-

10. What is Cross-Validation?

Answer:

Technique to evaluate model performance by splitting data into multiple folds and validating across them to avoid overfitting.

11. What is Regularization?

Answer:

Technique to prevent overfitting by penalizing large coefficients.

Types:

- L1 (Lasso)
 - L2 (Ridge)
-

12. Explain KNN Algorithm

Answer:

- Stores training data

- Classifies based on nearest neighbors
- Uses distance metrics like Euclidean

Limitation: High computation for large datasets

13. What is Decision Tree?

Answer:

Tree-based model that splits data using feature thresholds based on criteria like:

- Gini Index
 - Entropy
-

14. What is Random Forest?

Answer:

Ensemble of decision trees using:

- Bagging
- Feature randomness

Improves accuracy and reduces overfitting

15. What is PCA?

Answer:

Dimensionality reduction technique that transforms features into principal components capturing maximum variance.

3. ADVANCED LEVEL

16. What is Curse of Dimensionality?

Answer:

As features increase:

- Data becomes sparse
 - Model performance decreases
 - Computation increases
-

17. What are Ensemble Methods?

Answer:

- **Bagging** – Parallel models (Random Forest)
 - **Boosting** – Sequential learning (AdaBoost, XGBoost)
-

18. What is Generative vs Discriminative Model?

Answer:

- **Generative** → Models joint probability $P(X, Y)$ (e.g., Naive Bayes)
 - **Discriminative** → Models $P(Y|X)$ (e.g., Logistic Regression)
-

19. What is ROC-AUC?

Answer:

- ROC Curve → TPR vs FPR
 - AUC → Area under curve
 - Higher AUC = better model
-

20. What is Feature Engineering?

Answer:

Process of:

- Creating new features
- Transforming variables

- Encoding categorical data
-

21. Handling Imbalanced Data

Answer:

- Oversampling (SMOTE)
 - Undersampling
 - Class weights
 - Use F1 / ROC instead of accuracy
-

22. What is Pipeline in ML?

Answer:

Sequence of steps:

- Preprocessing
- Feature engineering
- Model training

Ensures reproducibility

23. What is Hyperparameter Tuning?

Answer:

- Grid Search
- Random Search
- Bayesian Optimization

Goal → Optimize model performance

24. Explain Reinforcement Learning

Answer:

Agent learns by interacting with environment using rewards and penalties instead of labeled data

4. EXPERT / SENIOR LEVEL

25. How do you choose the right ML algorithm?

Answer:

- Data size
 - Feature type
 - Interpretability
 - Latency requirements
 - Problem type (classification/regression)
-

26. How do you design an end-to-end ML system?

Answer:

1. Problem definition
 2. Data collection
 3. Data preprocessing
 4. Feature engineering
 5. Model selection
 6. Evaluation
 7. Deployment
 8. Monitoring
-

27. What is Model Drift?

Answer:

- Data distribution changes over time
- Causes performance degradation

Solution:

- Monitoring
- Retraining pipelines

28. Explain Online vs Batch Learning

Answer:

- **Batch Learning** → Train on entire dataset
- **Online Learning** → Incremental updates

29. What is Explainability in ML?

Answer:

Techniques:

- SHAP
- LIME
- Feature importance

Used for:

- Trust
- Compliance
- Debugging

30. How do you handle missing data?

Answer:

- Remove rows/columns
- Imputation (mean/median/mode)
- Predictive imputation

5. BONUS: SCENARIO-BASED QUESTIONS

31. Model performing well on training but poorly on test?

Answer: Overfitting → Apply regularization, cross-validation

32. Accuracy high but model useless?

Answer: Imbalanced dataset → Use F1, ROC-AUC

33. Too many features?

Answer: Use PCA / Feature selection

34. Training too slow?

Answer:

- Reduce features
- Use simpler model
- Parallel processing